

Biology — Detailed Marking Report

Federal Board Annual Examination 2026, Version 2 | Chapters 1-7 (Ch 7: Sec 7.1-7.6)

Candidate: Seemab | Attempt Date: 29 June 2026 | Marked: 29 June 2026 | Exam: exam-08-ch1-7

54.5/65
Total Score

83.8%
Percentage

A
Grade

Section A (MCQ): **12/12**

PERFECT — 8th consecutive

Section B (Short): **25/33**

Section C (Long): **17.5/20**

SECTION A — Multiple Choice Questions (12/12)

Q	Seemab's Answer	Correct Answer	Result
1	C	C	✓
2	C	C	✓
3	B	B	✓
4	C	C	✓
5	B	B	✓
6	C	C	✓
7	B	B	✓
8	B	B	✓
9	C	C	✓
10	C	C	✓
11	B	B	✓
12	C	C	✓

All 12 MCQs correct. This is Seemab's 8th consecutive perfect MCQ score, maintaining a 99.2% all-time MCQ record (250/252).

SECTION B — Short Answer Questions (25/33)

Q.2(i) OR Three careers in biology

3/3

Complete answer. Medicine/MBBS (diagnose and treat disease), Biotechnology (develop useful products), Agriculture (improve crop yield) — all three careers named with correct descriptions.

Q.2(ii) OR Role of Laveran and Ross in malaria

2/3

Ronald Ross (1897): fully correct. Fed female Anopheles mosquitoes on infected patients, dissected them, found Plasmodium in stomach walls, established the vector hypothesis.

Deduction –1: Alphonse Laveran (1882) entirely omitted. The question asks for both scientists. Laveran was the first to discover Plasmodium in the blood of a malaria patient and proposed that the parasite causes the disease.

Q.2(iii) OR Domain Archaea

2/3

Correctly explained that Archaea are prokaryotic, lack peptidoglycan in their cell walls, and inhabit extreme environments. The distinction from Bacteria was stated.

Deduction –1: The three types of archaeobacteria and their specific habitats are missing. Required: methanogens (swamps, gut of animals), halophiles (extremely salty environments), thermoacidophiles (hot acidic springs).

Q.2(iv) OR Chloroplast vs Mitochondria

3/3

Full comparison: double membrane (both); thylakoids/grana/stroma (chloroplast) vs cristae/matrix (mitochondrion); photosynthesis (chloroplast) vs ATP production (mitochondrion); plant cells only vs all eukaryotic cells. All marks earned.

Q.2(v) main Metaphase and Anaphase of mitosis

3/3

Both stages correctly described including the role of spindle fibres in alignment (Metaphase) and separation of sister chromatids to opposite poles (Anaphase). Complete.

Q.2(vi) main Prophase I of meiosis**3/3**

Synapsis, bivalent (tetrad), crossing over (non-sister chromatids, chiasmata, genetic variation) all defined correctly. Full marks.

Q.2(vii) main Homeostasis**1/3**

Homeostasis correctly defined as the maintenance of a stable internal environment.

Deduction –2: The entire homeostatic response sequence is missing. Required: stimulus → sensor (receptor) → control centre (e.g., brain/hypothalamus) → effector → response → negative feedback loop. Two marks are allocated to this mechanism and none of it was attempted.

Q.2(viii) main Shoot system vs Root system**2.5/3**

Shoot system fully described. Root system: location and primary/secondary root distinction correctly given.

Deduction –0.5: Functions of the root system missing. Required: absorption of water and minerals, anchoring the plant, and food storage.

Q.2(ix) OR Functions of lipids**2/3**

Four functions correctly stated: energy storage, cell membrane component, hormone synthesis, thermal insulation.

Deduction –1: The explanation of why lipids yield more energy per gram than carbohydrates is missing. Lipids have a lower oxygen-to-carbon ratio, so more energy is released on oxidation (approximately 9.3 kcal/g vs 4.1 kcal/g for carbohydrates).

Q.2(x) OR Nucleotide structure**1/3**

Correctly named the three components of a nucleotide: pentose sugar, nitrogenous base, phosphate group.

Deduction –2: The two types of pentose sugar and their assignment are missing. Required: deoxyribose (in DNA) and ribose (in RNA). This distinction is explicitly required by the marking scheme and is two marks.

Q.2(xi) OR Three types of cofactors**2.5/3**

All three cofactor types correctly defined: activator (inorganic, loosely bound), prosthetic group (organic, tightly/permanently bound), coenzyme (organic, loosely bound).

Deduction –0.5: No examples given for any of the three. The mark scheme requires at least one concrete example (e.g., Mg^{2+} / Zn^{2+} as activators; FAD/heme as prosthetic groups; NAD+/coenzyme A as coenzymes).

SECTION C — Long Answer Questions (17.5/20)**Q.3(i) OR** Three-domain system (Carl Woese, 1990)**4.5/5**

Strong answer overall. Basis of the system (rRNA sequence analysis, membrane lipid types, antibiotic sensitivity) clearly stated. Domain Archaea with all three types (methanogens, halophiles, thermoacidophiles); Domain Bacteria; Domain Eukarya with all four kingdoms (Animalia, Plantae, Fungi, Protista) — all present.

Deduction –0.5: Two factual errors within the Animalia section. (1) Animals described as "producers" — animals are consumers, not producers. (2) Vertebrates and invertebrates described in reverse — vertebrates have a backbone/vertebral column, invertebrates do not (not the other way around). These are factual slips in otherwise well-structured content.

Q.3(ii) main Structure and functions of the nucleus**3.5/5**

Good coverage: nucleus as control centre, DNA content, nuclear envelope (double membrane), nuclear pores (selective transport), nucleoplasm, nucleolus (rRNA and ribosome synthesis) — all correct.

Deduction –1.5: Three related items missing. (1) Chromatin: DNA wound around histone proteins forming chromatin fibres. (2) Chromatin condensing into chromosomes during cell division. (3) Gene and genome definitions — a gene is a segment of DNA coding for a protein; the genome is the entire DNA complement of a cell. The mark scheme allocates specific marks to these structural details.

Q.3(iii) OR Classification of carbohydrates**4.5/5**

Classification basis, general formula $C_x(H_2O)_y$ with 1:2:1 H:O ratio, monosaccharides ($C_6H_{12}O_6$, $C_5H_{10}O_5$) and disaccharides ($C_{12}H_{22}O_{11}$; sucrose, maltose, lactose with formation mechanism) all correct.

Deduction –0.5: Polysaccharides (the third carbohydrate type) entirely omitted. Required: examples such as starch (energy storage in plants), glycogen (energy storage in animals), and cellulose (structural role in plant cell walls).

Complete and accurate answer. Temperature: gradual rise increases activity to optimum (36–38 °C), denaturation of active site above optimum, activity resumes on cooling below optimum; inactive at very low temperatures. Substrate concentration: when enzyme is in excess, rate is directly proportional to substrate; rate plateaus when all active sites are saturated. All key terms (denaturation, optimum, saturation) correctly used and explained. Full 5/5.

DEDUCTION SUMMARY

Question	What was missing / wrong	Marks lost
2(ii) OR	Alphonse Laveran (1882) entirely omitted	–1
2(iii) OR	Three types of archaeobacteria + habitats missing	–1
2(vii) main	Homeostatic response sequence missing (only definition given)	–2
2(viii) main	Root system functions (absorption / anchoring / storage) missing	–0.5
2(ix) OR	"Why lipid yields more energy than carbohydrate" missing	–1
2(x) OR	Two pentose sugars (deoxyribose/ribose, which in DNA/RNA) missing	–2
2(xi) OR	Examples for the three cofactors missing	–0.5
3(i) OR	Animalia: "producers" instead of consumers; vertebrates/invertebrates reversed	–0.5
3(ii) main	Chromatin + chromatin→chromosomes + gene + genome definitions missing	–1.5
3(iii) OR	Polysaccharides (third carbohydrate type) omitted	–0.5
Total deducted		–10.5
Final score: 65 – 10.5 =		54.5

STRENGTHS AND AREAS FOR IMPROVEMENT

Strengths

- **Perfect MCQs (12/12):** 8th consecutive perfect score; all-time 250/252 (99.2%).
- **Organelle comparison (Q.2(iv)):** Chloroplast vs mitochondria fully covered with correct structural and functional details.
- **Mitosis stages (Q.2(v)):** Metaphase and Anaphase both correctly described including spindle role.
- **Meiosis Prophase I (Q.2(vi)):** Synapsis, tetrad, crossing over with chiasmata and genetic significance — all correct.
- **Enzyme question (Q.3(iv)):** Full 5/5; temperature optimum, denaturation, saturation — complete and precise.
- **Accuracy:** When Seemab does write something, it is almost always factually correct. The biology knowledge is solid.

Areas for Improvement

- **Stopping before all sub-parts:** 7 of 10.5 marks lost are correct openings left unfinished (Laveran, homeostasis sequence, pentose sugars, root functions, energy comparison).
- **Two-scientist questions:** Laveran/Ross — only one scientist addressed. Always check if a question names two people.
- **Homeostasis mechanism:** Definition alone is not enough. The stimulus→sensor→control centre→effector→negative-feedback chain must be written out.
- **Nucleotides:** Naming components is step one; identifying deoxyribose (DNA) vs ribose (RNA) is step two and must follow.
- **Factual slips in 3(i):** Animals are consumers, not producers. Vertebrates have a backbone; invertebrates do not.

Recurring Pattern (6th consecutive exam showing this)

Across the last six exams in multiple subjects, Seemab consistently starts questions correctly but stops before completing every required sub-part. This accounts for the majority of lost marks in this paper. Recommended drill before every question: read the question, list every sub-part asked, and tick each one off before moving on. This one habit change would likely add 6–8 marks to a typical paper.

Top Revision Targets

- **Homeostasis (Ch 5):** Memorise the full response sequence as a flow diagram: stimulus → receptor → control centre → effector → response → negative feedback.
- **Laveran and Ross (Ch 1):** Both dates and contributions. Laveran 1882 discovered Plasmodium; Ross 1897 identified mosquito as vector.
- **Nucleotide pentose sugars (Ch 6):** Ribose in RNA, deoxyribose in DNA — must be stated explicitly when nucleotides are discussed.
- **Polysaccharides (Ch 6):** Third class of carbohydrates (starch, glycogen, cellulose) with examples and roles.
- **Archaeobacteria types (Ch 2):** Methanogens, halophiles, thermoacidophiles with one habitat each.
- **Nucleus: chromatin and genome (Ch 3):** Chromatin = DNA + histones; chromatin condenses to chromosomes; gene = segment of DNA; genome = complete DNA set.

COMPARISON WITH PRIOR BIOLOGY ATTEMPTS

Date	Exam	Score	Percentage	MCQ	Grade
3 Apr 2026	exam-01-ch1-6	52/65	80.0%	11/12	A
30 Apr 2026	exam-02-ch7	60.5/65	93.1%	12/12	A
2 May 2026	exam-02-ch7 (re-sit)	65/65	100.0%	12/12	A+
8 Jun 2026	exam-03-ch4	62.5/65	96.2%	12/12	A+
23 Jun 2026	exam-07-ch1-7	57.5/65	88.5%	12/12	A
29 Jun 2026	exam-08-ch1-7 (this)	54.5/65	83.8%	12/12	A

83.8% is the second-lowest biology score overall (above only exam-01 at 80.0%). Among the two comprehensive ch1-7 papers, this is 4.7 percentage points below exam-07 (88.5%). Both incomplete-sub-part patterns should be addressed before the next full-syllabus paper.